

Cassava Leaf Disease: Catching the Rare Diseases

CS 412 Project Proposal

Team Members: Hannan Shahid, Yashashwi Singh
University of Illinois Chicago

1. Problem

Cassava is a major food crop in Africa, and leaf diseases have destroyed a large share of the harvest. Generally, an expert is required for Farmers to identify the disease. Our goal is to identify it automatically from a photo of a leaf.

We will use the Kaggle Cassava Leaf Disease dataset, which has 21,397 field photos from Uganda, each labeled as healthy, or one of four diseases. The biggest challenge for us is the imbalance of the diseases, because one disease (CMD) makes up about 61% of the images, while the rarest class is only about 5%. So a model will be able to reach 61% accuracy by always guessing CMD, which will never catch the other rare diseases.

2. Related Work, and Difference

Earlier work has used deep learning to detect plant and cassava diseases, and most have focused on raising the overall accuracy with CNNs such as EfficientNet. Our core paper proposed a class balanced loss that gives rare classes more weight during training. We can also use Focal loss as another approach to the same problem.

Overall accuracy hides whether a model catches the rare diseases, so we compare several models and several ways of handling imbalance, and we will report the results for each disease, not just the overall accuracy.

3. Plan

For this project we will be comparing different machine learning models to see which one of them can identify the cassava leaf diseases. The models we will be comparing is:

1. We will first be using a classic model using HOG and color features with an SVM.
2. We will then use and compare two CNN models, ResNet-50 and EfficientNet-B0.

After we are done with comparing the models, we will use the best model to test the different ways of handling the imbalance in the dataset. For instance:

1. Normal training without making any changes for the imbalance.
2. Oversampling where the images from the rare disease classes are shown more often during the training.

3. Class balanced loss which gives more importance to the classes that have fewer/lesser images.
4. Focal loss which helps the model focus more on examples that are harder to classify/organize.

Then to compare the results we will look at the accuracy, marco f1 score, recall for each disease, and confusion matrices. This will help us understand not only how accurate each of the models is overall, but also how it can identify each of the diseases, especially the rare ones.

We will be using python and pytorch for the project and google colab to train the models and all of our code and progress will also be kept/be on github as mentioned on the rubrics.

4. Milestones

Date	Milestone
October 1	Finishing the proposal and setting up the github repository
October 2 to October 14	Loading and understand the dataset and completing the classical baseline model
October 15 to October 31	Training and testing ResNet-50 and EfficientNet-B0
November 1	Completing the intermediate project report
November 2 to November 14	Testing the different ways of handling the imbalance and comparing the results for each disease
Mid November	Preparing and completing the project presentation and leaf photo demo
December 1	Competing and submitting the final project report

Team Roles

Hannan will work on the different ways of handling the imbalance, including oversampling, class balanced loss, and focal loss. Hannan will also analyze the results for each of the diseases.

Yashashwi will be working on preparing the dataset and the classical baseline model as well as training the ResNet-50 and EfficientNet80-B0 models.

Both of us will be working together on comparing the final results, preparing the presentation, and writing the intermediate and final reports.

References

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